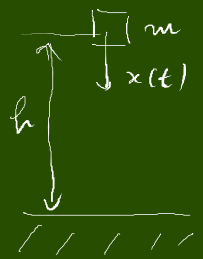
Ecuații diferențiale

modelare ⟶ rezolvare ⟶ interpretare



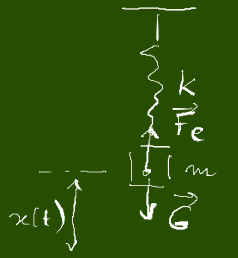
$ma = mg$

$a = g \qquad x'' = g$

$x(t) \longrightarrow x'(t) \longrightarrow x''(t)$

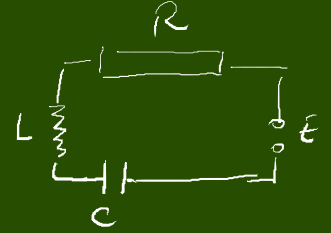
$v(t) \qquad a(t)$

$\begin{cases} t & \text{variabilă} \\ x & \text{fct. nec.} \end{cases}$



RLC

$ma = mg - kx \Rightarrow x'' + \frac{k}{m}x = g$



$I(t)$ intensitatea curentului

$U_R = I(t) \cdot R$

$U_C = \frac{Q(t)}{C}$

$Q(t)$? $I(t)$

$I(t) = \frac{\Delta Q}{\Delta t} \left(= Q'(t) \right)$

$$U_L = L \cdot I'(t)$$

$$\Rightarrow \quad R \cdot I + \frac{\int_0^t I(s)\,ds}{C} + L I' = E$$

$$Q$$

$$L \cdot Q'' + R \cdot Q' + \frac{1}{C} Q = E$$

Ecuație diferențială de ordinul $\overline{I}$ $\qquad x' = F(t, x)$ (variabilă, fct. necunoscută $x = x(t)$)

$F : I \times D \longrightarrow \mathbb{R}$ cont.

$I \subseteq \mathbb{R}$ interval

$D \subseteq \mathbb{R}^n$

Rezolvarea ec : $x = x(t)$ derivabilă $\quad x : J \subseteq I \longrightarrow D$

$$x'(t) = F(t, x(t)) \quad \forall\, t \in J$$

Problema Cauchy IVP
$$\begin{cases} x' = F(t, x) \\ x(t_0) = x_0 \end{cases} \qquad t_0 \in I \,,\ x_0 \in D$$

(initial value problem)

Thm. (fundam.)

$F : I \times D \longrightarrow \mathbb{R}$ $\quad \exists \frac{\partial F}{\partial x}$, $\frac{\partial F}{\partial x}$ cont. (F este o fct. de clasă C^1 în raport cu x)

$\Rightarrow \quad \forall t_0 \in I$, $\forall x_0 \in D \quad \exists!\ x : J \subseteq I \longrightarrow D$ a.î.

$$\begin{cases} x'(t) = F(t, x(t)) & \forall t \in J \\ x(t_0) = x_0 \end{cases}$$

Ecuația dif. liniară de ordinul I $\quad x' = \underbrace{f(t)x + g(t)}_{F(t,x)}$

$f, g : I \subseteq \mathbb{R} \longrightarrow \mathbb{R}$ cont.

I $\quad \boxed{x' = \lambda x}$ $\quad \lambda \in \mathbb{R}$

$\lambda = 1 \quad x' = x \quad x'(t) = x(t)$

$x(t) = e^t$

($x' \leftarrow \frac{dx}{dt} \leftarrow \dot{x} \leftarrow x_t$ $\quad x'' \leftarrow \frac{d^2x}{dt^2} \leftarrow \ddot{x} \leftarrow x_{tt}$)

$\lambda \in \mathbb{R}$ $\qquad x' = \lambda x$

$x(t) = e^{\lambda t}$ Sol.

$x'(t) = \lambda x(t) \quad | \cdot e^{-\lambda t}$

$x'(t) e^{-\lambda t} - \lambda e^{-\lambda t} x(t) = 0 \Rightarrow x(t) e^{-\lambda t} = C \qquad C \in \mathbb{R}$

$\underbrace{x'(t) e^{-\lambda t} - \lambda e^{-\lambda t} x(t)}_{(x(t) e^{-\lambda t})'}$

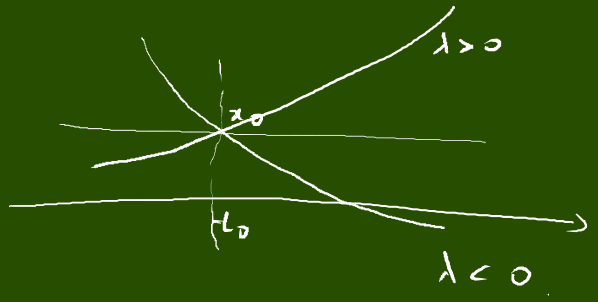
$\boxed{x(t) = C e^{\lambda t}}$

Pb. Cauchy $\begin{cases} x' = \lambda x \\ x(t_0) = x_0 \end{cases}$ $\Rightarrow$ $x_0 = C e^{\lambda t_0} \Rightarrow C = x_0 e^{-\lambda t_0}$

$\Rightarrow \boxed{x(t) = x_0 \cdot e^{\lambda (t - t_0)}}$

$\lambda > 0 \quad x \uparrow$

$\lambda < 0 \quad x \downarrow$



exemple. $x' = \lambda x$ — creșterea populației (T. Malthus)

$\lambda = 1.05\%$ pe an

$t_0 = 2021 \quad x(t_0)$

$t_1 = ? \quad x(t_1) = 2x(t_0)$

$x(t_1) = x(t_0) \cdot e^{\lambda(t_1 - t_0)}$

$2x(t_0)$

$\Rightarrow \lambda(t_1 - t_0) = \ln 2$

$t_1 = t_0 + \frac{\ln 2}{0{,}0105} \simeq t_0 + 66$

II $x' = f(t)x$, $f : I \subseteq \mathbb{R} \to \mathbb{R}$ cont.

$F(t)$ primitivă pt f $\quad F'(t) = f(t)$

a) $x(t) = e^{F(t)}$ sol. a ec. dif. $x' = f(t)x$

$e^{F(t)} \cdot F'(t) = f(t) e^{F(t)}$ ✓

b) $x' = f(t)x$ $\quad x'(t) = f(t)x(t) \quad | : e^{F(t)}$

$x'(t)e^{-F(t)} - f(t)e^{-F(t)}x(t) = 0$

$$\left(x(t)e^{-F(t)}\right)' = 0$$

$$\Rightarrow \quad x(t)e^{-F(t)} = C$$

$$\boxed{x(t) = C \cdot e^{F(t)}}, \quad C \in \mathbb{R}.$$

Obs. 1) $S_0 = \{ x : I \longrightarrow \mathbb{R} \mid x' = f(t)x, \ x \text{ derivabilă} \}$

sp. vectorial (principiul superpoziției)

$\dim_{\mathbb{R}} S_0 = 1$ bază $\{e^{F(t)}\}$

G altă primitivă

$G(t) = F(t) + D, \ D \in \mathbb{R}$

$\Rightarrow$ Sol. gen. $x(t) = Ke^{G(t)} =$

$= K \cdot e^{F(t)} e^{D}$

$= Ce^{F(t)}$

2) pb. Cauchy $\begin{cases} x' = f(t)x \\ x(t_0) = x_0 \end{cases}$

$F(t) = \int_{t_0}^{t} f(s)\,ds$ $\qquad F'(t) = f(t)$, $F(t_0) = 0$.

$x(t) = C e^{\int_{t_0}^{t} f(s)\,ds}$, $x(t_0) = x_0 \Rightarrow x_0 = C \cdot e^{0}$

$\Rightarrow \boxed{x(t) = x_0 e^{\int_{t_0}^{t} f(s)\,ds}}$

III $x' = f(t)x + g(t)$ $\qquad f, g : I \subseteq \mathbb{R} \longrightarrow \mathbb{R}$ cont.

Metoda variației constantei

Pas 1 $\qquad x' = f(t)x \Rightarrow x_0(t) = Ce^{F(t)}$, $C \in \mathbb{R}$

$F'(t) = f(t)$

Pas 2 $\qquad x' = f(t)x + g(t)$

Caut sol. de forma $x(t) = C(t)e^{F(t)}$ unde $C(t)$ este o fct

pe care o voi determina

$C'(t)e^{F(t)} + C(t)e^{F(t)} \cdot f(t) = f(t)C(t)e^{F(t)} + g(t)$

$$c'(t) = g(t)e^{-F(t)} \quad \Rightarrow \quad c(t) = \int g(t)e^{-F(t)}\,dt + k, \quad k \in \mathbb{R}$$

$\Rightarrow$ soluția generală a ec. diff. $x' = f(t)x + g(t)$ este

$$x(t) = e^{F(t)}\left(\int g(t)e^{-F(t)}\,dt + k\right), \quad k \in \mathbb{R}$$

Pb. Cauchy $\begin{cases} x' = f(t)x + g(t) \\ x(t_0) = x_0 \end{cases}$

$$F(t) = \int_{t_0}^{t} f(s)\,ds \qquad c(t) = \int_{t_0}^{t} g(s)e^{-F(s)}\,ds$$

$$\boxed{x(t) = e^{\int_{t_0}^{t} f(s)\,ds}\left(\int_{t_0}^{t} g(s)e^{-\int_{t_0}^{s} f(z)\,dz}\,ds + k\right)}$$

$x(t_0) = x_0 \quad \Rightarrow \quad x_0 = k$

Exemple: 1) $\begin{cases} (1+t^2)x' = 2tx + 1 + t^2 \\ x(0) = -2 \end{cases}$

$$x' = \underbrace{\frac{2t}{1+t^2}}_{f(t)} x + \underbrace{1}_{g(t)}$$

$x' = \frac{2t}{1+t^2} x \Rightarrow x_0(t) = C e^{F(t)}$, $F(t) = \int f(t)\,dt = \int \frac{2t}{1+t^2}\,dt =$

$= \ln(1+t^2)$

$x_0(t) = C e^{\ln(1+t^2)} = C(1+t^2)$

$x' = \frac{2t}{1+t^2} x + 1$ $\quad x(t) = C(t)(1+t^2)$

$C'(t)(1+t^2) + C(t) \cdot 2t = \frac{2t}{1+t^2} \cdot C(t) \cdot (1+t^2) + 1$

$\Rightarrow C'(t) = \frac{1}{1+t^2} \Rightarrow C(t) = \int \frac{1}{1+t^2}\,dt = \operatorname{arctg} t + k, \; k \in \mathbb{R}$

$\Rightarrow x(t) = (1+t^2)(\operatorname{arctg} t + k)$

$x(0) = -2 \Rightarrow -2 = k$

$\Rightarrow \quad x(t) = (1+t^2)(\operatorname{arctg} t - 2)$

2) A $\qquad T(t) \searrow \qquad T' = -k(T-A) \qquad k > 0$

$T' = -kT + kA$

$T' = -kT \quad \Rightarrow \quad T(t) = e^{-kt} \cdot C$

$T' = -kT + kA$

$T(t) = e^{-kt} C(t) \quad \Rightarrow$

$\Rightarrow \quad -ke^{-kt}C(t) + e^{-kt}C'(t) = -k \cdot e^{-kt}C(t) + kA$

$C'(t) = kAe^{kt}$

$C(t) = Ae^{kt} + B \quad , B \in \mathbb{R}$

$\Rightarrow \quad T(t) = e^{-kt}(Ae^{kt} + B) = A + Be^{-kt} \xrightarrow{t \to \infty} A$

Aplicație : $\quad A = 24^\circ C \qquad t_1 = 8{:}00 \text{ AM} \qquad T(t_1) = 28^\circ C$

$t_2 = 9{:}00 \text{ AM} \qquad T(t_2) = 26^\circ C$

$t_0 = ? \qquad T(t_0) = 37^\circ C$

$$\begin{cases} 24 + Be^{-8k} = 28 \\ 24 + Be^{-9k} = 26 \end{cases} \qquad Be^{-8k} = 4 \qquad \frac{Be^{-9k} = 2}{e^k = 2} \Rightarrow k = \ln 2$$

$$\Rightarrow B = 4 \cdot e^{8k} = 4e^{8\ln 2} = 2^{10}$$

$$T(t) = 24 + 2^{10} \cdot e^{-t\ln 2} = 24 + 2^{10-t} \searrow 24$$

$$T(t_0) = 37 \Rightarrow 2^{10-t} = 13$$

$$10 - t = \log_2 13 \simeq 3{,}5$$

$$t \simeq 6{,}5 \qquad 6:30 \text{ AM}$$

3) $x' = \underline{\lambda x - \mu x^2} \qquad \lambda, \mu > 0$

ecuația logistică

$$\lambda x - \mu x^2 = 0 \begin{cases} x(t) = 0 \\ x(t) = \frac{\lambda}{\mu} \end{cases}$$

$x(t)$ $\qquad y(t) = \frac{1}{x(t)} \Rightarrow x(t) = \frac{1}{y(t)}$

$$-\frac{y'(t)}{y^2(t)} = \lambda \cdot \frac{1}{y(t)} - \mu \cdot \frac{1}{y^2(t)} \quad \Big| (-y^2(t))$$

$$y'(t) = -\lambda y(t) + \mu .$$

$$y' = -\lambda y + \mu . \quad \Rightarrow \quad y(t) = e^{-\lambda t} \cdot C(t)$$

$$C'(t) e^{-\lambda t} = \mu \quad \Rightarrow \quad C(t) = \frac{\mu e^{\lambda t}}{\lambda} + \frac{k}{\lambda} = \frac{\mu e^{\lambda t} + k}{\lambda}$$

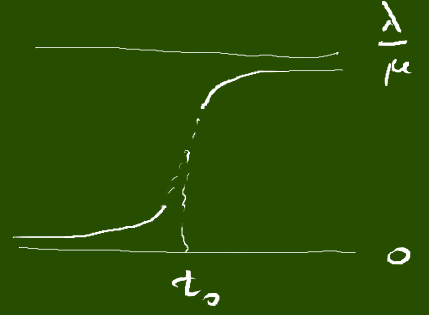
$$\Rightarrow \quad y(t) = e^{-\lambda t} \frac{(\mu e^{\lambda t} + k)}{\lambda} = \frac{\mu + k e^{-\lambda t}}{\lambda}$$

$$\Rightarrow \quad x(t) = \frac{\lambda}{\mu + k e^{-\lambda t}} , \quad k \in \mathbb{R}$$

fcd. logarítmica

$$t \longrightarrow -\infty \qquad x(t) \longrightarrow 0$$

$$t \longrightarrow +\infty \qquad x(t) \longrightarrow \frac{\lambda}{\mu} .$$



$$\lambda = \mu = k = 1 \qquad x(t) = \frac{1}{1+e^{-t}} = \frac{e^t}{1+e^t} = \frac{1}{2} + \frac{1}{2} \tanh\left(\frac{t}{2}\right)$$

SIR